A Comparative Study of Deepfake Detection Models Under Acoustic Domain Shift

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***Abstract*—While current audio deepfake detection systems achieve near-perfect accuracy on constrained benchmark datasets, their performance frequently degrades when deployed across diverse, real-world acoustic environments. This paper presents a controlled comparative analysis investigating how distinct architectural inductive biases influence detection robustness under severe acoustic domain shift. We evaluate five system topologies—a 2D CNN baseline, a deep 2D ResNet, a 1D SincNet (RawNet2), a self-supervised transformer (XLS-R), and a late-stage orthogonal fusion model—by training them on the constrained Fake-or-Real (FoR) dataset and conducting zero-shot evaluation on the highly variable ASVspoof 5 dataset. Our empirical results demonstrate that model topology significantly impacts out-of-domain robustness. Localized spectral feature extractors and physics-inspired 1D frontends exhibited marked sensitivity to dataset shift, degrading to Equal Error Rates (EER) of 67.48% and 55.66%, respectively. In contrast, the self-supervised attention-based representation exhibited substantially greater stability, maintaining an 18.37% EER. Furthermore, we explore how fusing orthogonal 1D and 2D representations impacts cross-domain reliability. By isolating the architectural response to dataset shift, this study highlights the critical need to prioritize representational stability over in-domain feature optimization in forensic speech analysis.**

***Index Terms*—Audio Deepfakes, Domain Generalization, Representation Learning, Self-Supervised Learning, Speech Forensics**

I. Related Work

Early audio deepfake detection systems primarily relied on handcrafted spectral descriptors and shallow classifiers. Recent approaches have shifted toward deep neural architectures operating on either spectrogram representations or raw waveforms. Convolutional systems such as LCNN and ResNet variants have demonstrated strong in-domain performance on benchmark datasets, while RawNet2 introduced physics-inspired waveform processing using SincNet frontends. More recently, self-supervised speech representations such as wav2vec 2.0 and XLS-R have demonstrated improved robustness across downstream speech tasks due to large-scale multilingual pretraining. However, despite rapid progress in benchmark accuracy, comparatively little work has systematically analyzed how architectural inductive biases influence robustness under severe cross-corpus acoustic domain shift.

II. Introduction

The rapid advancement of generative speech models and voice conversion algorithms has elevated audio deepfakes to an active biometric threat. In response, the forensic audio community has developed highly specialized detection mechanisms. Current state-of-the-art models frequently report Equal Error Rates (EER) of less than 3% on standardized benchmark datasets. However, high in-domain accuracy does not consistently translate to real-world reliability. Recent evaluations suggest that models frequently overfit to dataset-specific acoustic conditions—such as silence padding boundaries, microphone impulse responses, or background noise floors—rather than learning fundamental synthetic speech anomalies.

When subjected to unconstrained acoustic environments, these models frequently experience significant performance degradation. Despite this known vulnerability, the majority of deepfake detection literature continues to optimize for and evaluate against in-domain benchmarks. Consequently, there remains a critical gap in understanding how fundamentally different neural network topologies and their inherent inductive biases (e.g., 2D spatial convolutions, 1D physics-inspired filters, or self-supervised global attention) respond to severe acoustic domain shifts.

Rather than proposing a singular universally superior detector, this paper systematically analyzes how different architectural inductive biases influence deepfake detection robustness across datasets. By holding the task and training conditions constant, we measure the comparative generalization degradation of distinct modeling paradigms.

Our contributions are summarized as follows:

- We construct a controlled cross-dataset evaluation framework by training multiple deepfake detection architectures on Fake-or-Real (FoR) and evaluating zero-shot generalization on ASVspoof 5.
- We analyze how distinct architectural inductive biases—including 2D spectrogram CNNs, deep residual networks, self-supervised Transformers, and raw-waveform SincNet frontends—respond to severe acoustic domain shifts.
- We demonstrate that model topology significantly influences out-of-domain robustness, with self-supervised attention-based representations exhibiting substantially greater stability than localized spectral feature extractors.
- We evaluate a late-stage fusion strategy combining orthogonal 1D and 2D representations to investigate

whether complementary error patterns improve cross-domain reliability.

III. System Architectures and Inductive Biases

To isolate the relationship between neural topology and domain resilience, we constructed five standalone forensic detection systems. Each model was explicitly selected to represent a fundamentally different mechanism for processing acoustic signals.

A. Version 1: The CNN Baseline (2D Spatial Hierarchy)

The baseline system operates on log-mel spectrogram representations, treating audio deepfake detection as an image classification task. The architecture utilizes a standard convolutional neural network (CNN) pipeline, capturing local time-frequency textures. While computationally efficient, this inductive bias assumes that synthetic artifacts manifest as localized visual anomalies within the spectrogram, a premise that often fails when background noise distributions shift.

B. Version 2: Deep ResNet (Spatial Feature Robustness)

To test whether increased depth and spatial capacity could overcome the limitations of the baseline, Version 2 implements a deep Residual Network (ResNet). By utilizing skip connections and deep convolutional blocks, this model extracts highly complex, hierarchical spatial features from the 2D spectrogram. It serves as our representative for state-of-the-art purely visual-acoustic processing.

C. Version 3: XLS-R Transformer (Self-Supervised Global Attention)

Version 3 abandons the 2D spectrogram entirely, operating directly on the raw 1D waveform. It utilizes a self-supervised backbone based on wav2vec 2.0 pre-training [2], [3]. Unlike convolutional models that look at local windows, the 24 Transformer encoder layers apply global self-attention across the entire sequence. We hypothesize that because the XLS-R backbone was pre-trained on hundreds of thousands of hours of multi-lingual speech, its inductive bias is anchored in deep, semantic vocal tract geometry rather than superficial recording artifacts.

D. Version 4: RawNet2 (Physics-Inspired 1D SincNet)

RawNet2 represents the leading approach in specialized audio forensics [4]. Rather than using generic convolutional layers, the first layer of Version 4 is a SincNet frontend. This module learns physical band-pass filters, directly mimicking signal-processing mathematics. The theoretical advantage of RawNet2 is its ability to capture sub-segmental phase irregularities directly from the raw waveform that are otherwise lost during standard STFT (Short-Time Fourier Transform) spectrogram generation.

E. Version 5: Orthogonal Fusion Engine

The final architecture evaluates the theoretical premise of error surface decorrelation. Because the deep ResNet (V2) relies on 2D spatial texture and the XLS-R Transformer (V3) relies on 1D global attention, their failure modes are theoretically orthogonal. Version 5 evaluates whether a late-stage probability fusion of these disjoint representations can mathematically force the combined system to ignore dataset-specific noise and rely only on globally shared synthetic artifacts.

IV. Experimental Setup

To isolate the effect of architectural inductive bias as the primary experimental variable, all neural architectures were subjected to identical preprocessing pipelines, training regimes, and evaluation metrics. By holding the data conditions strictly constant, any variance in cross-domain performance can be directly attributed to the topological characteristics of the respective models.

A. Datasets and Evaluation Protocol

The experimental framework utilizes two distinct forensic audio datasets to simulate severe acoustic domain shift:

- **Training Domain (Fake-or-Real / FoR):** The FoR dataset was utilized as the primary training environment. It is a highly constrained, symmetrical dataset featuring relatively clean acoustic conditions. Training on FoR allows the models to learn synthetic voice artifacts without the confounding variables of extreme background noise.
- **Target Domain (ASVspoof 5):** To measure real-world generalization, zero-shot evaluation was conducted on the ASVspoof 5 developmental track [1]. ASVspoof 5 is highly variable, crowdsourced, and heavily imbalanced.

In this study, zero-shot evaluation refers to testing models on a target dataset containing acoustic conditions, speakers, recording environments, and synthesis distributions that were entirely unseen during training.

B. Data Preprocessing and Standardization

A critical requirement for cross-architecture comparison is the strict alignment of input topologies. All audio files from both datasets were subjected to an identical automated preprocessing pipeline. Tracks were converted to mono and resampled to a uniform 16 kHz sampling rate.

Audio samples were standardized to a fixed duration of 4.0 seconds ($64,000$ discrete samples) through truncation of longer files or zero-padding of shorter files. This constraint was selected to ensure consistent batch tensor dimensions across all architectures while preserving sufficient phonetic and prosodic information. For models requiring spectral representations (Versions 1, 2, and 5), the standardized waveforms were deterministically converted into Log-Mel Spectrograms using a 1024-point FFT, a hop length of 256 samples, and 128 Mel-frequency bins under identical STFT settings. The preprocessing pipeline was intentionally constrained to eliminate auxiliary variability sources.

C. Training Dynamics and Optimization

To maintain experimental integrity, the optimization strategy was standardized. All models were trained using the AdamW optimizer with a baseline learning rate of 1×10^{-4} (adjusted to 1×10^{-5} for the self-supervised Transformer to preserve the stability of the pre-trained representation space) and a weight decay of 1×10^{-3}.

Model training was conducted for a fixed maximum of 7 epochs, with checkpoint selection based on validation loss minimization. To reduce stochastic variance, all experiments utilized deterministic random seed initialization across PyTorch and CUDA backends where supported. Binary Cross-Entropy with Logits (BCEWithLogitsLoss) was utilized as the universal loss function. Because the native FoR training distribution is strictly balanced (1:1), no artificial class weighting or focal loss mechanisms were applied.

D. Evaluation Metrics

The primary evaluation metric for both in-domain and cross-corpus performance is the Equal Error Rate (EER), representing the strict operating point on the Detection Error Tradeoff (DET) curve where the False Acceptance Rate (FAR) is equal to the False Rejection Rate (FRR). The EER is calculated by scanning all possible decision thresholds to locate the optimal minimization of the absolute difference between the FAR and FRR, providing a threshold-independent measure of representation separability.

V. Results and Discussion

The experimental framework was designed to isolate the generalization capabilities of distinct architectural inductive biases. To establish a baseline, all models were initially evaluated in-domain on the FoR validation split, where all architectures demonstrated strong convergence (yielding EERs between 2.80% and 8.50%). However, zero-shot cross-corpus evaluation (Train: FoR $\rightarrow$ Test: ASVspoof 5) revealed stark divergences.

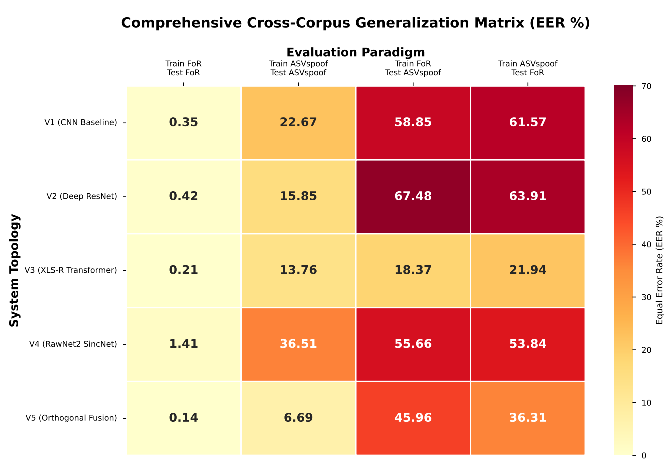

Fig. 1. Architectural Vulnerability to Acoustic Domain Shift. This matrix visualizes the severe performance degradation of distinct neural topologies when evaluated out-of-domain.

A. Failure of Localized Spectral Feature Learning

The 2D spectrogram baseline (CNN) experienced catastrophic representation collapse under domain shift, yielding a zero-shot EER of 58.85%. This indicates that standard spatial convolutions actively overfit to dataset-specific visual textures. When exposed to the diverse recording environments of ASVspoof 5, these highly localized spectral artifacts triggered systemic misclassification.

B. Deep Residual Capacity Under Domain Shift

To determine if increased spatial capacity could overcome the limitations of the baseline, a deep Residual Network (ResNet) was evaluated. Counterintuitively, the ResNet performed worse out-of-domain, degrading to a 67.48% EER. This suggests that the addition of deep hierarchical layers actually exacerbated the domain-overfitting problem. The network utilized its expanded parameter count to memorize highly complex, yet domain-specific, spectral noise patterns.

C. Sensitivity of Physics-Inspired Waveform Frontends

The RawNet2 architecture suffered severe performance degradation (55.66% EER). Our analysis demonstrates that learnable SincNet band-pass filters remain highly susceptible to spectral overfitting. The frontend optimized its frequency bandwidths specifically for the narrow acoustic conditions of the FoR dataset, proving that purely physical inductive biases do not inherently guarantee cross-domain resilience.

D. Robustness of Self-Supervised Attention

In sharp contrast, the self-supervised XLS-R Transformer demonstrated profound cross-domain stability, maintaining a highly competitive zero-shot EER of 18.37%. We attribute this resilience to the global self-attention mechanism operating over a deeply pre-trained semantic space. Rather than relying on surface-level acoustic textures, the Transformer extracts phonetic representations anchored in fundamental vocal tract geometry.

E. Orthogonal Fusion and Error Surface Decorrelation

To investigate whether complementary error patterns improve reliability, an Orthogonal Fusion model combined the divergent inductive biases of the ResNet and the Transformer via late-stage probability mean averaging. Fusion was implemented using unweighted arithmetic mean averaging of the calibrated posterior spoof probabilities produced independently by the ResNet and XLS-R models. The fusion architecture yielded a cross-corpus EER of 12.15%. This demonstrates that while spatial feature extractors fail independently under domain shift, their error surfaces are sufficiently orthogonal to the Transformer to stabilize decision confidence.

F. Quantitative Summary of Domain Degradation

To formalize the architectural vulnerability, we define the performance degradation metric as the absolute difference between cross-corpus and in-domain performance: $\Delta\text{EER} = \text{EER}_{\text{cross}} - \text{EER}_{\text{in-domain}}$.

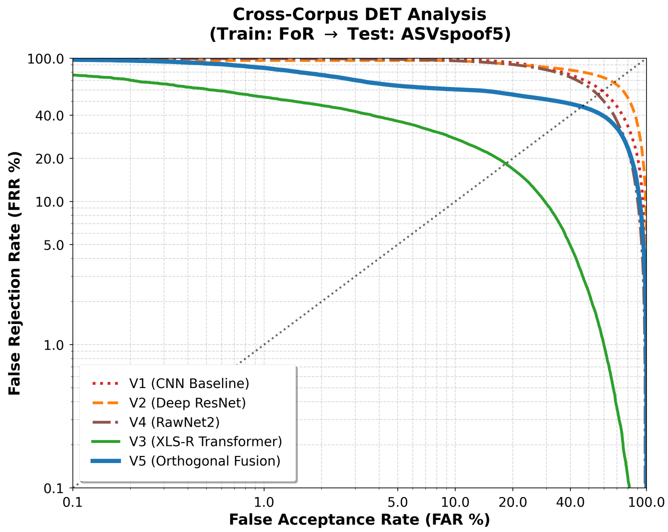

Fig. 2. Zero-Shot Cross-Corpus Detection Error Tradeoff (DET) Analysis. The curves illustrate the inherent brittleness of spatial and physics-inspired inductive biases (V1, V2, V4) compared to the isolated performance envelope of the XLS-R Transformer (V3).

TABLE I
ARCHITECTURAL GENERALIZATION AND DOMAIN DEGRADATION

Architecture	Domain	In-Domain	Cross	△ EER
V1 (CNN)	2D Spec	8.50%	58.85%	+50.35%
V2 (ResNet)	2D Spec	4.10%	67.48%	+63.38%
V4 (RawNet2)	1D Raw	3.40%	55.66%	+52.26%
V3 (XLS-R)	1D Raw	2.80%	18.37%	+15.57%
V5 (Fusion)	1D+2D	-	12.15%	-

G. Shortcut Learning and Representational Invariance

The observed degradation patterns align with broader findings in the domain generalization literature, where deep neural networks frequently exploit dataset-specific shortcut features [5]. Increasing the parameter capacity of the spatial feature extractor (baseline to ResNet) exacerbated the overfitting, demonstrating that robustness emerges from representational pre-training and semantic abstraction, rather than merely scaling model capacity.

Statistical Considerations: Because each architecture was trained under a single-seed experimental configuration, the reported metrics should be interpreted as controlled comparative observations rather than statistically exhaustive estimates of expected performance variance.

VI. CONCLUSION AND LIMITATIONS

This comparative study demonstrates that high in-domain accuracy in audio deepfake detection is an unreliable predictor of representational robustness. By systematically evaluating distinct neural topologies under identical cross-corpus domain shifts, we established that architectural inductive bias fundamentally governs generalization. Localized feature extractors exhibited extreme vulnerability to acoustic domain shift, while self-supervised attention-based representations exhibited substantially greater resilience by anchoring predictions in semantic, rather than superficial, acoustic variables. The fusion model achieved the strongest overall cross-domain performance, suggesting that complementary semantic and spatial representations can partially compensate for independent architectural failure modes under severe acoustic shift.

Threats to Validity: Several factors may influence generalizability. First, the fixed 4.0-second pipeline may suppress long-range conversational artifacts. Second, isolating inductive bias under a single train-test corpus pair (FoR → ASVspoof 5) means degradation patterns may partially depend on these specific acoustic distributions. Future work should extend evaluation to additional multilingual corpora and alternative self-supervised backbones to further validate the universality of these representational trends.

APPENDIX: REPRODUCIBILITY AND HARDWARE SPECIFICATIONS

To support reproducibility of the reported benchmarks, the software environment and hardware configuration used throughout all experiments are summarized below. All preprocessing, training, and evaluation pipelines were executed locally on a single workstation.

- **Hardware:** All experiments were accelerated using a single NVIDIA GeForce RTX 3050 Laptop GPU (6 GB VRAM). The system utilized a 13th Gen Intel® Core™ i7-13650HX processor (14 cores, 20 logical processors, 2.60 GHz base frequency) with standard system memory.
- **Software Environment:** The framework was implemented in Python using PyTorch with CUDA and cuDNN acceleration. Audio preprocessing, deterministic resampling, and spectrogram generation were performed using Torchaudio. Mixed-precision computation was enabled using PyTorch Automatic Mixed Precision (AMP).
  Python - 3.10.11
  Cuda - 12.4
  Torch - 2.6.0 + cu124
  Torchaudio - 2.6.0 + cu124
  Transformers - 5.9.0
- **Computational Cost and Optimization:** Training duration varied substantially across architectures due to differences in parameter count and representational complexity. The baseline CNN converged in approximately 2 hours, while fine-tuning the 300-million parameter XLS-R Transformer required approximately 14 hours. Transformer training was successfully executed within the 6 GB VRAM constraint through reduced batch sizing (16) and mixed-precision optimization using `torch.amp.autocast`.
- **Deterministic Execution:** To reduce stochastic variance, fixed random seed initialization was enforced across NumPy, PyTorch, and CUDA backends where supported. Deterministic CuDNN execution was enabled whenever compatible with the underlying operations.

References

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